

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SUBJECT: Foundations of Computer Security

Mid-Semester Examination

■ CS 6301 ■ Duration 120 minutes ■ Full Marks 30 ■ Date: 07-October-2023■

Answer ALL Questions

All parts of the question must be answered at one place.

Number within square bracket indicate full mark for that question.

1. (a) What is DES? A Feistel cipher is used in the DES algorithm. Describe the operation of a Feistel cipher. What are the Weaknesses of Data Encryption Standard? [3]
(b) A message has 3000 characters. If it is to be encrypted using a block cipher of 128 bit, find the size of padding and number of blocks. [2]
(c) Show an LFSR with the characteristics polynomial $x^4 + x + 1$. What is the period? [2]
(d) Design a polybious key matrix for play fair cipher using the keyword "EFFECTIVENESS" and decrypt the message CUVQNWZBGNPMQVIEWGVCEXZBAT. [3]
2. (a) What are the various components used in modern symmetric key encipherment? Discuss each with neat diagram. [3]
(b) Encrypt the first 8 characters of your name using a Autokey cipher and the key is the last digit of your roll number. Start with your first name, and if it is less than 8 characters continue with your second name or surname, ignore spaces. [2]
(c) Define Z, Z^*, Z_p, Z_p^* . [2]
(d) Find the multiplicative inverse of the matrix given below over Z_{37} . [3]
$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$
3. (a) Use the extended Euclidean algorithm to find the multiplicative inverse of the polynomial $x^5 + x^2 + 1$ in $GF(2^7)$ using the modulus $x^7 + x + 1$. [2]
(b) Name three different families of algebraic groups that are commonly used in cryptographic applications and explain their properties. [2]
(c) The encryption key in the transposition cipher is (6,4,3,1,5,2). Find the decryption key. [2]
(d) Use a brute-force attack to decipher the following message. Assume that you know it is an Affine cipher and that the plain text "ab" is enciphered to "LS". [4]
OINLYHJNAOF0INTVNHOPFYYVRSNAOJFPH

*****Wish you all the best*****